

11 (long)

(a) Figure 9 shows *nominal stress – nominal strain* curves I, II, III, IV obtained from tensile tests for four alloys A, B, C and D described in Table 1. Identify which curve corresponds to which alloy, explaining your reasoning and discussing the reasons for the differences in the curves. [8]

(b) A ductile wire with a cross-section initially uniform is loaded in tension. At the maximum load, necking is observed to start. A criterion for the onset of necking is

$$\frac{d\sigma_t}{d\varepsilon_t} = \sigma_t$$

where σ_t is the true stress and ε_t is the true strain. The relationship between true stress σ_t and true strain ε_t for an annealed alloy wire undergoing a tensile test approximates to

$$\sigma_t = 250 \varepsilon_t^{1/2} \text{ MPa}$$

Determine:

- (i) the true strain in the wire when the tensile load reaches its maximum value; [4]
- (ii) the tensile strength of the alloy. [6]

(c) Alloy wire samples, with tensile strength σ_{ts} , are subjected to an alternating stress about a zero mean stress. Under these test conditions a stress range of $0.50\sigma_{ts}$ gives a fatigue life of 10^6 cycles. Repeating the test with a stress range of $0.65\sigma_{ts}$ reduces the fatigue life to 10^4 cycles. In a second series of tests, a stress range of $0.50\sigma_{ts}$ is applied at a mean stress of $\sigma_m = 0.10\sigma_{ts}$ for 4×10^4 cycles. The stress range is then increased to $0.55\sigma_{ts}$ and testing is continued until failure occurs. Estimate the total fatigue life of the samples in the second series of tests. [12]

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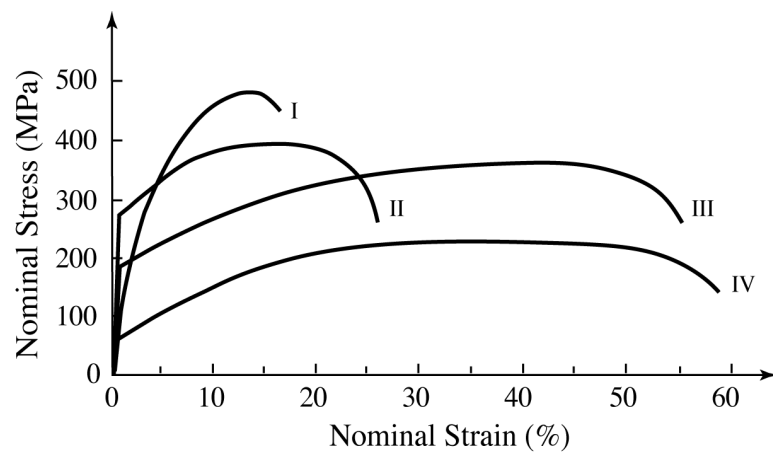


Fig. 9

A	Brass, annealed
B	Brass, drawn
C	Heat-treatable Al alloy
D	Copper, annealed

Table 1